

VELMÈRE ADVANCED AUDIT REPORT

Bitcoin Core (Native Layer-1 Network (No EVM Contract · UTXO))

VELMERE SECURITY AUDIT REPORT: BITCOIN CORE

Audited Contract: Native Layer-1 Network (No EVM Contract · UTXO)

Network: Bitcoin Mainnet (Chain ID: 0 (UTXO Mainnet))

Report ID: vlm_btc_advanced_263 | Tier: ADVANCED | Application Surface: BROWSER

Created At: 2026-09-07T21:38:32.531Z

VERDICT SUMMARY: NOT SCORED (MISSING BYTECODE)

Confidence Score: 0/100 | Evidence Coverage: 0%

Bytecode is missing or malformed. In accordance with Velmère truth invariants, bytecode-derived claims are NOT emitted and security risk is marked NOT SCORED.

--- PROTOCOL OVERVIEW & NATIVE L1 NETWORK CONTEXT [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of distributed architecture and consensus mechanism

Protocol / Asset: Bitcoin Core

Architecture Type: Native Base Layer (Layer-1 · No Smart Contract)

Network Identifier: native_chain:btc

Native Symbol: BTC

Rule Governance: Distributed node consensus across validators / miners

--- CONSENSUS SPECIFICATION & NODE CLIENT VERIFICATION [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Analysis of block validation rules, transactions, and cryptographic curves

Verification of Bitcoin Core is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT_APPLICABLE for native base protocol assets.

--- BASELINE NETWORK SECURITY FINDINGS [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified risk vectors in base layer architecture

--- CONSENSUS GOVERNANCE & PROTOCOL UPGRADE ANALYSIS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Decentralization evaluation, validator distribution, and improvement proposal process

Network Consensus Governance: Distributed Node Consensus [VERIFIED]

Smart Contract Admin Keys: NOT_APPLICABLE (Native Layer-1 Protocol) [VERIFIED]

Emergency Network Pause Capability: NOT_APPLICABLE (Decentralized Network) [VERIFIED]

Arbitrary User Balance Drain: NOT_APPLICABLE (Cryptographic Keys Enforced) [VERIFIED]

Protocol rules for Bitcoin Core are enforced by distributed node consensus; no privileged smart contract owner key exists (NOT_APPLICABLE).

--- SPOT MARKET DEPTH, EXCHANGE RESERVES & NETWORK VELOCITY (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Global market liquidity, supply velocity, and mempool dynamics

Market Reserve Liquidity: Global Spot Market Liquidity [VERIFIED]

DEX AMM Pairs: NOT_APPLICABLE (Native Base Asset) [NEUTRAL]

Transfer Tax / Sell Fee: NOT_APPLICABLE (Standard Network Fee Model) [VERIFIED]

Liquidity for Bitcoin Core is anchored in global spot venues and custodial settlement networks; automated market maker liquidity locks are NOT_APPLICABLE.

--- NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Modeling of chain reorganizations, Sybil attacks, and P2P partitioning

VELMÈRE ADVANCED AUDIT REPORT

Bitcoin Core (Native Layer-1 Network (No EVM Contract · UTXO))

External Oracle Dependency: NOT_APPLICABLE (Protocol Native Settlement) [VERIFIED]
Flash Loan Attack Vector: NOT_APPLICABLE (Non-EVM Execution Environment) [VERIFIED]
Double-Spend Attack Vector: Secured by network consensus [VERIFIED]

--- STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED) [ADVANCED] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Binary compatibility verification and state transition engine determinism
State Transition Determinism: Conforms to protocol specification [VERIFIED]
Client Implementation Diversity: Reference implementations evaluated [VERIFIED]
State integrity for Bitcoin Core is enforced by deterministic state transitions; EVM storage layout diffing is NOT_APPLICABLE.

--- PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED) [ADVANCED] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Tracking of consensus rule evolution and backward compatibility guarantees
Protocol Status: Active Production Mainnet
Consensus Fuzzing: NOT EXECUTED (Engine not commissioned)

--- HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED) [ADVANCED] ---
VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW
Attestation by independent distributed systems auditor
Review status: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED. In accordance with Velmère policy, human review claims are NEVER emitted unless verified reviewer evidence exists.
Reviewer State: INDEPENDENT HUMAN REVIEW NOT COMMISSIONED
Analyst: NONE (Independent manual review not commissioned)
Attestation Signature: NONE (No attestation emitted - review not performed)

CONFIDENTIALITY & LEGAL NOTICE:

This document is an evidence-bound security analysis report produced by Velmere Security.
It does not represent a commercial warranty, investment advice, or guaranteed-safe certification.